
HOUSE PRICE PREDICTION - Summary Report

Student: Harshit Dwivedi | Date: June 20, 2026 | Dataset: Housing Prices

OBJECTIVE: Build regression model to predict house prices and identify key influencing features.

KEY FINDINGS

1. MOST INFLUENTIAL FEATURES

- Area (square footage) | Bathrooms | Air conditioning

2. MODEL PERFORMANCE

Linear Regression: $R^2 = 0.6529$ (65.29%) | MAE: \$970,043

Random Forest: $R^2 = 0.6076$ (60.76%) | MAE: \$1,028,076

3. DATA INSIGHTS

- 545 properties | Average: \$4,766,729
- Right-skewed distribution
- Linear relationships dominate

4. DISCOVERY

Air conditioning significantly impacts value. Linear Regression outperformed Random Forest (65.29% accuracy).

RECOMMENDATION

- Use ML for pricing (79.6% error reduction)
 - Focus on area, bathrooms, air conditioning
 - Target \$3.43M-\$5.74M price range
-
-

METHODOLOGY

Data: 545 rows, one-hot encoding, no missing values

Models: Linear Regression, Random Forest (100 trees)

Evaluation: MAE, RMSE, R^2

Visualizations: Distribution, correlation, predictions

CONCLUSION

Machine learning achieves 65.29% accuracy in price prediction.

Area, bathrooms, and air conditioning are critical factors for real estate pricing decisions.